Data Center Project Report

Course: IT 401 - Data Centers

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1 Executive Summary

This report documents the design, simulation, and validation of a comprehensive network infrastructure for a three-story enterprise facility. The project aims to support approximately 600 users across five distinct departments (Sales, HR, Finance, Research, and Student Services) while ensuring high availability, security, and scalability.

Key architectural features include a hierarchical design utilizing Core, Distribution, and Access layers, dual-ISP redundancy for fault tolerance, and centralized network management. The implementation leverages VLANs for logical segmentation, OSPF for dynamic routing convergence, DHCP for automated address management, and SNMP for proactive system monitoring.

2 Network Architecture & Topology

The network topology was designed following the Cisco three-tier hierarchical model to ensure modularity and ease of maintenance. The infrastructure spans three physical locations: the First Floor, Second Floor, and the Underground Data Center.

2.1 Topology Overview

As illustrated in Figure 1, the design incorporates redundancy at the Core layer to prevent single points of failure. Two Multilayer Switches act as the distribution aggregation points, routing traffic between the departmental VLANs and the Core Routers.

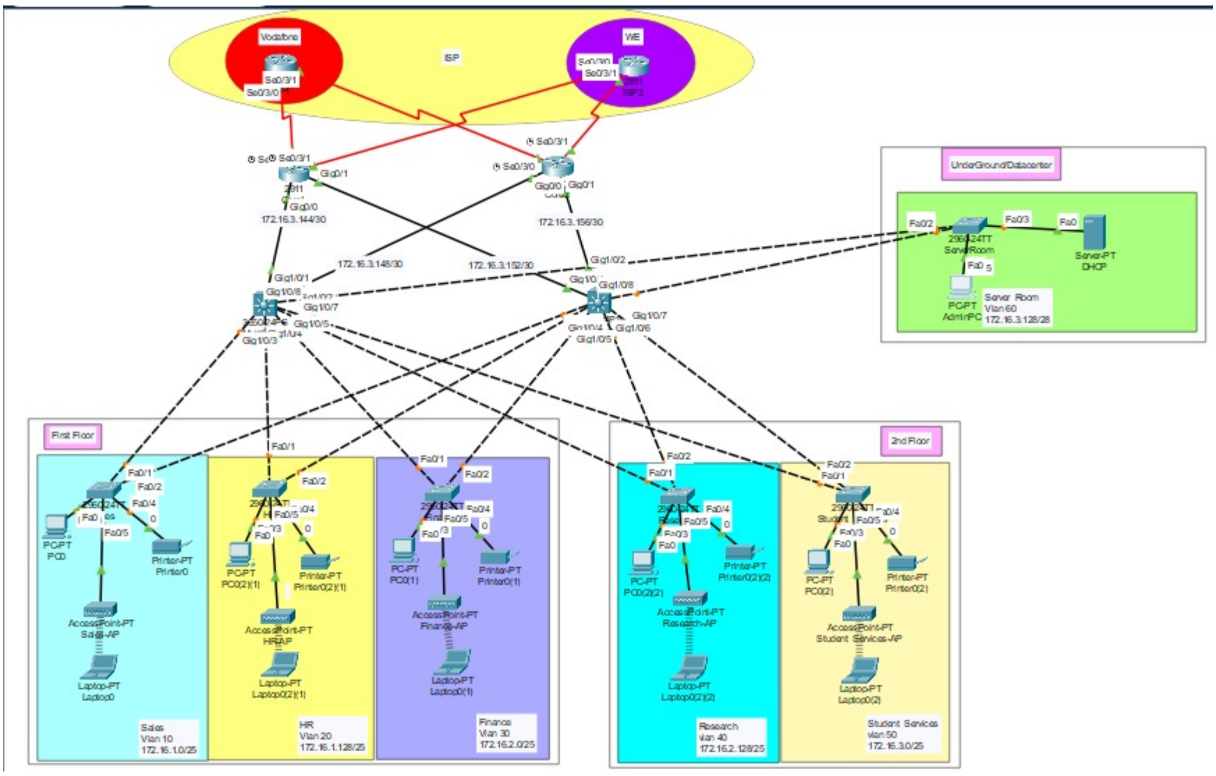

Figure 1: Complete Enterprise Network Topology Simulation (Packet Tracer)

3 IP Addressing and Subnetting

To optimize the usage of the allocated ‘172.16.1.0/22‘ address space, Variable Length Subnet Masking (VLSM) was employed. This approach allows for subnets to be sized according to the specific user density of each department, minimizing wasted addresses.

Department	VLAN ID	Network Address	CIDR	Usable Hosts
Sales	10	172.16.1.0	/25	126
HR	20	172.16.1.128	/25	126
Finance	30	172.16.2.0	/25	126
Research	40	172.16.2.128	/25	126
Student Svcs	50	172.16.3.0	/25	126
Server Room	60	172.16.3.128	/28	14

Table 1: VLSM IP Addressing Scheme

4 Implementation and Verification

4.1 VLAN Configuration and Segmentation

To reduce broadcast domains and enhance security, traffic was segmented using Virtual Local Area Networks (VLANs). Each department is isolated within its own broadcast domain. Inter-VLAN routing is handled by the Multilayer Switches using Switched Virtual Interfaces (SVIs).

Figure 2 demonstrates the verification of the VLAN database on ‘Multilayer1‘, confirming that all required VLANs (10, 20, 30, 40, 50, 60) are active and correctly named.

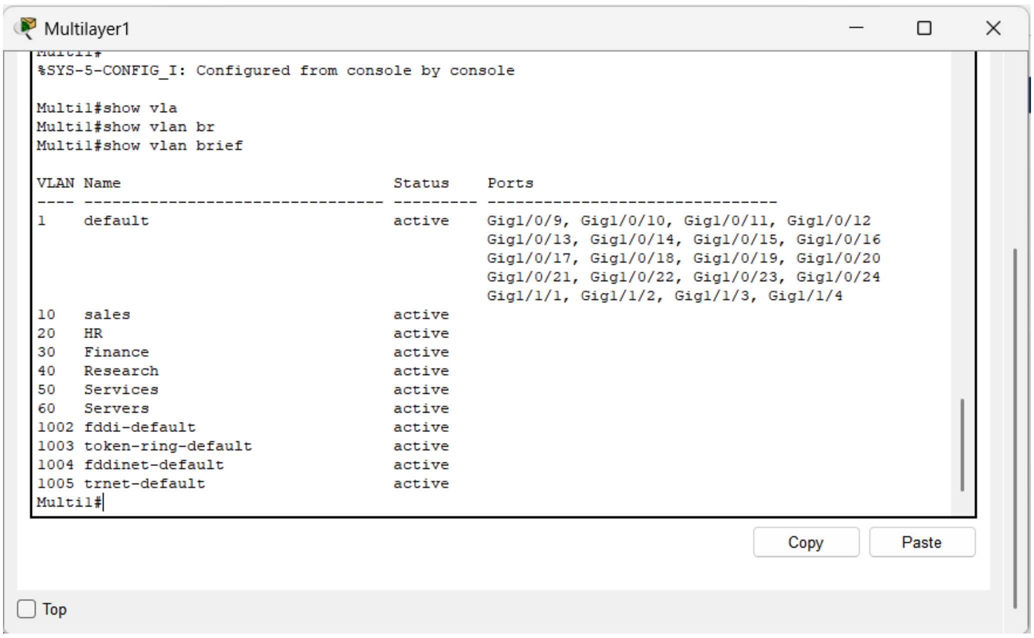

Figure 2: Verification of VLAN Database on Distribution Layer Switch

4.2 Dynamic Routing Protocols (OSPF)

Open Shortest Path First (OSPF) was selected as the interior gateway protocol due to its fast convergence and support for VLSM. All routers and multilayer switches were configured in **Area 0** (Backbone Area).

4.2.1 Routing Table Verification

The routing table on the Multilayer Switch (Figure 3) confirms that the switch has successfully learned routes to the external networks and other internal subnets via OSPF (indicated by the 'O' flag).

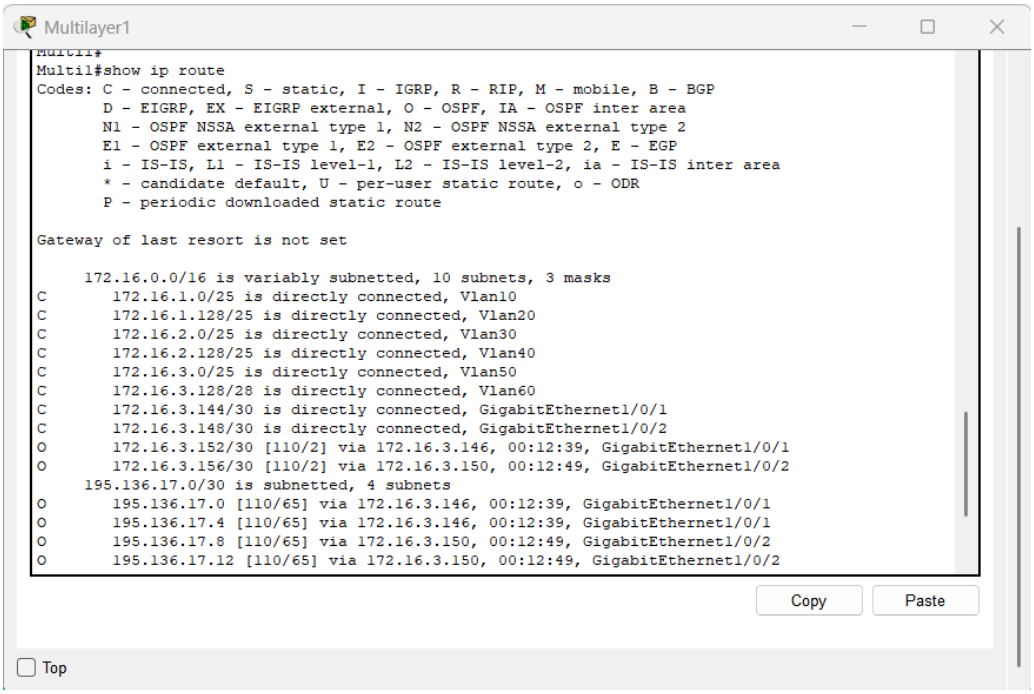

Figure 3: Multilayer Switch Routing Table showing OSPF Routes

4.2.2 Redundancy and Neighbor Adjacencies

To ensure high availability, both Core Routers established full adjacencies with the ISP routers and the internal Multilayer switches. Figure 4 and Figure 5 below display the 'show ip ospf neighbor' output, verifying that the neighbor relationships are in the 'FULL' state, which is critical for load balancing and failover.

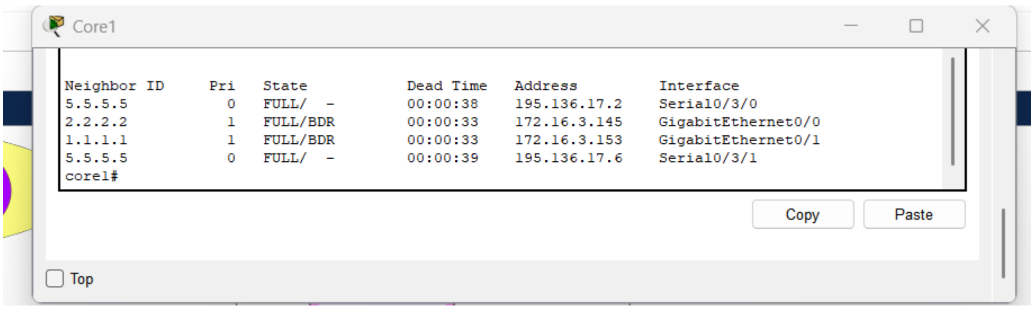

Figure 4: OSPF Neighbor Adjacency Verification on Core Router 1

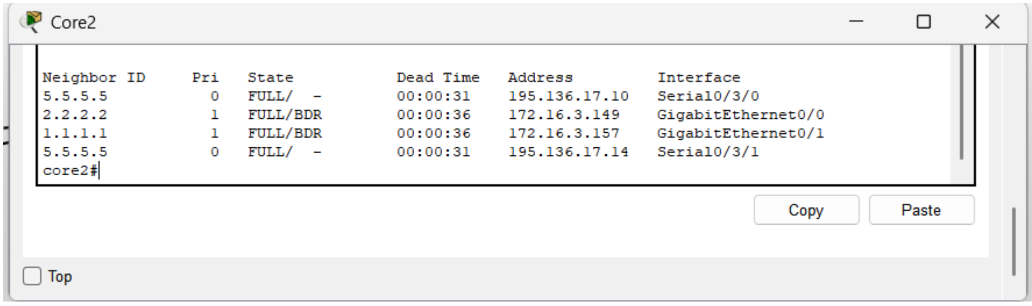

Figure 5: OSPF Neighbor Adjacency Verification on Core Router 2

4.3 Dynamic Host Configuration Protocol (DHCP)

A centralized DHCP server is deployed in the Server Room (VLAN 60) to manage IP allocation dynamically. DHCP Relay Agents ('ip helper-address') were configured on the Distribution layer SVIs to forward broadcast requests from departmental VLANs to the central server. Figure 6 illustrates the server configuration, highlighting the distinct pools created for each department.

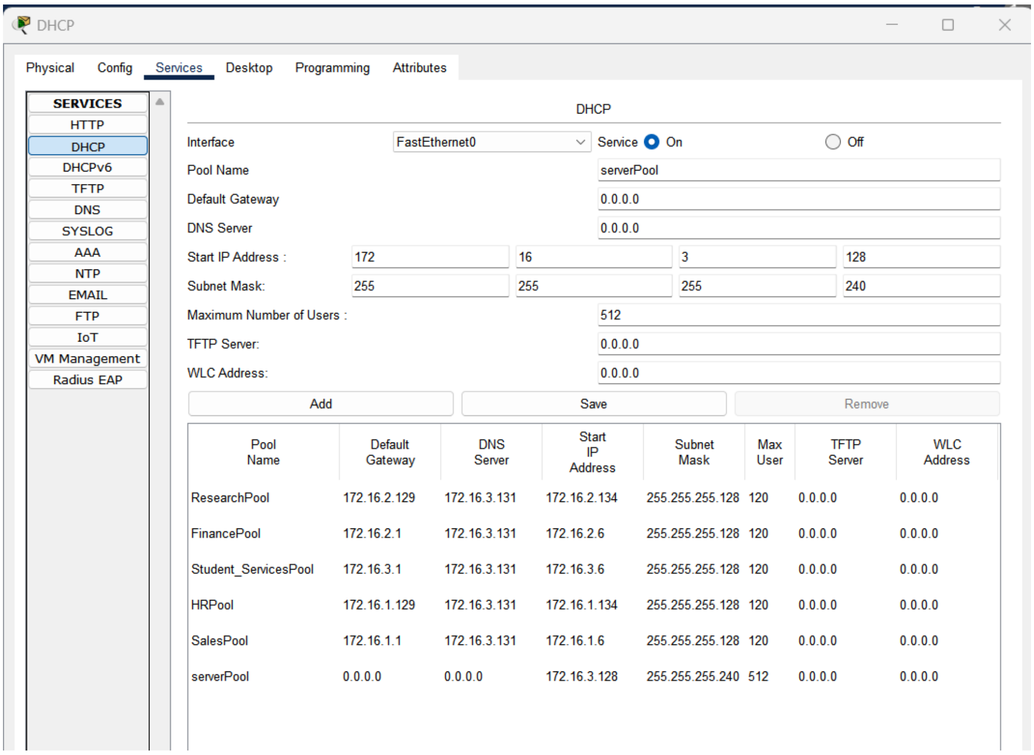

Figure 6: DHCP Server Pool Configuration and Status

4.4 Network Management (SNMP)

To enable proactive monitoring, Simple Network Management Protocol (SNMPv2c) was implemented. The Admin PC acts as the Network Management Station (NMS). Using a MIB Browser, we successfully queried the OID for 'sysName' on the network devices. Figure 7 shows the successful retrieval of device data, confirming that the Read-Only community string ('public') is correctly configured and the management plane is functional.

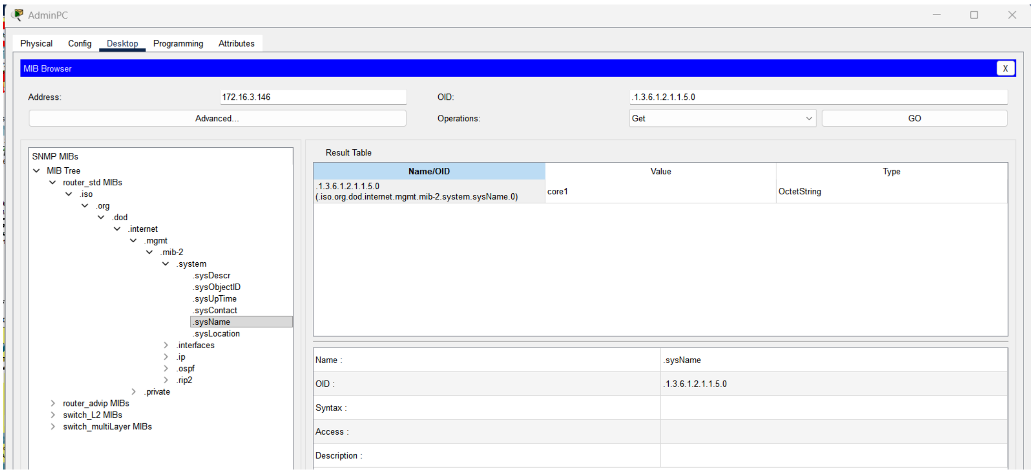

Figure 7: SNMP MIB Browser Test Results (Admin PC)

5 Conclusion

The Enterprise Network Design project was successfully implemented and simulated. The network meets all specified requirements, including:

- **Scalability:** The hierarchical design and VLSM addressing support future growth.
- **Redundancy:** OSPF and dual ISPs provide robust failover capabilities.
- **Manageability:** Centralized DHCP and SNMP monitoring simplify administrative overhead.

The validation tests confirm that all departments have full connectivity, security policies are in place, and the infrastructure is ready for deployment.